3.C Re-sampling / Re-weighting logistic regression.

Let n_p be # of positive examples in D.
Let n_n be # of negative examples in D.
Let n be total # of examples in D.

Fraction of positive examples in D is computed as

$$p = \frac{n_p}{n_p + n_n} = \frac{n_p}{n}.$$

$$\text{Given } K = \frac{p}{1-p} \Rightarrow K = \frac{\frac{n_p}{n}}{\frac{n_n}{n}} = \frac{n_p}{n_n}$$

$$\therefore \boxed{n_p = K n_n} \longrightarrow (1)$$

$$\text{Balanced accuracy} = \frac{1}{2}\left(A_0 + A_1\right)$$

$$\text{where } A_1 \triangleq \frac{TP}{TP+FN}, \quad A_2 \triangleq \frac{TN}{TN+FP}$$

$$\text{Balanced Accuracy of } D = \frac{1}{2}\left(\frac{TP}{n_p} + \frac{TN}{n_n}\right)$$

Now focusing on D'

Let n'_n be no. of negative examples in D'.

$$\text{Given } \boxed{n_n = n'_n} \longrightarrow (2)$$

Let n'_p be no. of positive examples in D'.

$$\text{Given } \boxed{n'_p = \frac{n_p}{K} = \frac{\cancel{K}\cdot n_n}{\cancel{K}} = n_n} \rightarrow (3)$$

(from (1))

Let n' be total number of samples in D'.

3.c

$$n' = n'_p + n'_n$$

from ② & ③, $n' = n_n + n_n = 2n_n$.

Accuracy is defined as

$$A = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Accuracy of } D' = \frac{TP' + TN'}{n'} = \frac{TP' + TN'}{n'_n + n'_p}$$

$$\text{Accuracy of } D' = \frac{TP' + TN'}{2n_n} \quad (\text{from ② \& ③})$$

$$= \frac{1}{2}\left[\frac{TP' + TN'}{n_n}\right] = \frac{1}{2}\left[\frac{TP'}{n_n} + \frac{TN'}{n_n}\right]$$

Since Given that for every positive example in D, there are $1/k$ repetitions in D' and the fact that our model and parameters haven't changed b/w D and D' then we can confidently say

$$TP' = \frac{TP}{k}.$$

with similar logic, we can confidently say

$$TN' = TN$$ (every negative sample appeared once and no change in model)

$$\therefore \text{Accuracy of } D' = \frac{1}{2}\left[\frac{TP}{k \cdot n_n} + \frac{TN}{n_n}\right]$$

from ① $k \cdot n_n = n_p$

$$\text{Accuracy of } D' = \frac{1}{2}\left[\frac{TP}{n_p} + \frac{TN}{n_n}\right]$$

3.c Accuracy of $D' = \frac{1}{2}\left[\frac{TP}{n_p} + \frac{TN}{n_n}\right]$

we already derived that the

Balanced Accuracy of $D = \frac{1}{2}\left[\frac{TP}{n_p} + \frac{TN}{n_n}\right]$

As the values we derived of Accuracy of D' and Balance Accuracy of D, we can confirm that

Balanced Accuracy of D = Accuracy of D'.